

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I T. Karimulla / 1924650 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T. Karimulla

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Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.
2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.
3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.
4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.
6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.
7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.
8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.
9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and

the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Given

File size = 150 MB

Segment size = 1500 bytes

Header + Trailer = 40 bytes

Bandwidth = 50 Mbps

Calculation:

File size = $150 \times 1,000,000 = 150,000,000$ bytes

Number of segments = $150,000,000 \div 1500 = 100,000$ segments

Number of frames = 100,000 frames

Total overhead = $100,000 \times 40 = 4,000,000$ bytes (4 MB)

Total transmitted data = $150 \text{ MB} + 4 \text{ MB} = 154 \text{ MB}$

Transmission time = $(154 \times 8) \div 50 = 24.64$ seconds

Explanation:

- Transport layer divides data into segments and ensures reliable delivery using acknowledgements and error recovery.
- Data link layer adds headers/trailers, detects errors using CRC and delivers frames reliably over the physical link.

② Given

• Window size = 6

Total frames = 48

Transmission windows = $48 \div 6 = 8$ windows

One frame lost in each window

Retransmission = 8 frames

Explanation:

Flow control prevents the sender from overwhelming the receiver, reducing congestion and improving communication efficiency.

③

Given

Packet = 12000 bytes

MTU = 1500 bytes

IP header = 20 bytes

Maximum payload per fragment = $1500 - 20 = 1480$ bytes

Fragments = $12000 \div 1480 = 9$ fragments

Header overhead = $9 \times 20 = 180$ bytes

Explanation:

The Network Layer fragments packets to fit the MTU.

The destination reassembles the fragments using identification and fragment offset fields.

④

Windows = 8

Retransmission = 8 frames

Flow control: It matches transmission speed with receiver capacity, avoiding packet loss and improving efficiency.

File = 600 MB

checkpoint every 60 MB

Failure after 390 MB

Checkpoints created = $390 \div 60 = 6$ checkpoints

Data transmitted = $390 - 360 = 30$ MB

Explanation:

Synchronization checkpoints allow communication to resume from the last checkpoint instead of restarting from the beginning

⑥ Given

Original file = 900 MB

Compression = 40%

Encrypted overhead = 5%

Compressed size = $900 \times 0.60 = 540$ MB

Encrypted size = $540 \times 1.05 = 567$ MB

Reduction = $((900 - 567) \div 900) \times 100 = 37\%$

Explanation:

The presentation layer compresses data to reduce transmission time and encrypts due to provide Confidentiality and security,

⑦ Given

- Data = 30 MB
- Throughput = 120 Mbps
- Request size = 3 MB

Convert data = 3000 MB

Request = $3000 \div 3 = 1000$ requests

Transmission time = $(3000 \times 8) \div 120 = 200$ seconds

Explanation:

The Application layer provides services such as file transfer, authentication, and user interaction while ensuring reliable communication.

⑧ Processing delays = $4 + 3 + 5 + 2 + 1 = 15$ ms

Propagation delay = 18 ms

Transmission delay = 12 ms

Total delay = $15 + 18 + 12 = 45$ ms

Explanation:

- Application layer processes user requests
- Transport layer ensures reliable communication
- Network layer routes packets
- Data link layer frames data and performs error detection
- Physical layer transmits bits over the medium

Explanation:

The presentation layer compresses data, the transport layer divides it into segments, the data link layer adds frame overhead, and the physical layer transmits the final data reliably and securely.

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Criteria	Understanding of concepts	Explanation	Application	Organization
	25%	25%	20%	15%

Accuracy	clarity
10 %	5 %